

SAYANA JOY

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SUMMARY

Software Engineer with 5 years of experience building scalable, user-focused web applications. Experienced in delivering performant, accessible, and full-stack solutions through cross-functional collaboration.

KEY SKILLS

- **Technologies** - JavaScript, React.js, Redux, TypeScript, Jest, HTML/CSS, SCSS, Tailwind CSS
- **Backend & Cloud** - Node.js, PostgreSQL, SQL, AWS, CI/CD Pipelines
- **Accessibility & Tools** - WCAG 2.1, JAWS, NVDA, axe DevTools, Git, GitHub, Postman, Jira

PROFESSIONAL EXPERIENCE

Systems Engineer, Tata Consultancy Services, Kerala, India

November 2022–July 2024

- Contributed to the development of an enterprise-scale React application built for a Fortune 500 U.S. financial services client, collaborating within a cross-functional frontend team of three.
- Implemented WCAG 2.1-compliant accessibility features (JAWS, NVDA) across shared UI components, while maintaining 80%+ Jest test coverage to ensure build stability and production readiness.
- Reviewed frontend merge requests, mentored a junior engineer, and resolved 20+ post-release production issues, reducing critical bug turnaround time by 40%.
- Led performance optimization efforts by reducing unnecessary re-renders, implementing memoization strategies, ensuring cross-browser compatibility, and reducing API calls, resulting in faster load times and smoother interactions.

Senior Software Engineer, Vanilla Networks Pvt. Ltd, Kerala, India

December 2020–November 2022

- Led the introduction and implementation of D3.js for advanced data visualization.
- Translated Figma designs into reusable, production-ready UI components aligned with consistent design patterns.
- Designed and implemented dynamic AG Grid-based data tables to support scalable, interactive data rendering.
- Designed and built scalable, reusable UI components within a shared component library, reducing duplicate code and improving consistency across 10+ features and Agile releases using React, TypeScript, React Context, and Emotion CSS.

Software Engineer, Attinad Software, Kerala, India

December 2019–November 2020

- Developed consumer-facing React UI components using SCSS, Bootstrap, and Redux, including WebSocket driven chat functionality and Stripe-integrated payment flows, improving checkout usability and overall user experience.
- Created interactive data visualizations using ApexCharts to enhance business reporting dashboards.
- Conducted 8+ client demos, collaborating with stakeholders to refine UI features based on feedback.

ACADEMIC EXPERIENCE

Teaching Assistant, University of Houston, TX

August 2025–May 2026

- Mentored 100+ undergraduate students through assignment reviews, technical guidance, and weekly office hours while independently leading full class sessions to ensure effective instruction and continuity of coursework.

PROJECTS

CodePractice - AI-Powered Full-Stack Interview Preparation Platform

May 2026

- Architected and built an AI-powered coding interview platform using React, TypeScript, Redux Toolkit, Node.js, Express, PostgreSQL, and OpenRouter AI, enabling real-time code execution and intelligent complexity analysis.
- Designed and implemented scalable backend APIs and PostgreSQL relational schemas supporting categorized coding challenges, hidden test case validation, dynamic execution workflows, and AI-driven complexity analysis across 5+ test cases per problem.
- Engineered a responsive multi-panel coding workspace using Tailwind CSS, Monaco Editor, and Redux Toolkit, delivering interactive execution feedback and optimized rendering performance.

Gender Classification from Text, University of Houston, TX

May 2025

- Built a gender classification ML pipeline using BERT embeddings, readability features, and K-Means clustering, achieving 74.3% accuracy with an ensemble of SVM, Random Forest, and XGBoost models.

EDUCATION

University of Houston, TX - MS in Computer Science

May 2026

APJ Abdul Kalam Technological University, India - B.Tech in Computer Science

July 2019